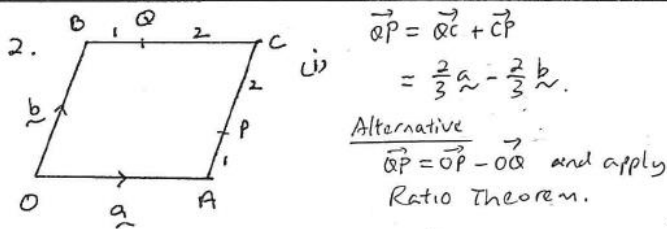
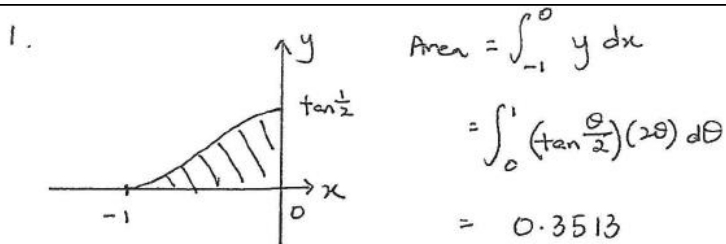


Victoria Junior College
H2 Mathematics (9740) – 2012 Preliminary Examinations P1 Solutions



ii) $\vec{OC} \cdot \vec{QP} = (\vec{a} + \vec{b}) \cdot \frac{2}{3}(\vec{a} - \vec{b})$

$$= \frac{2}{3}(\vec{a} \cdot \vec{a} - \vec{b} \cdot \vec{b}) = \frac{2}{3}|\vec{a}|^2 - \frac{2}{3}|\vec{b}|^2$$

iii) $\vec{OC} \cdot \vec{QP} = 0 \Rightarrow |\vec{a}| = |\vec{b}|$

$\therefore$ OACB is a rhombus.

3i) $\frac{1}{a+bx} = (a+bx)^{-1} = a^{-1} \left(1 + \frac{b}{a}x\right)^{-1}$

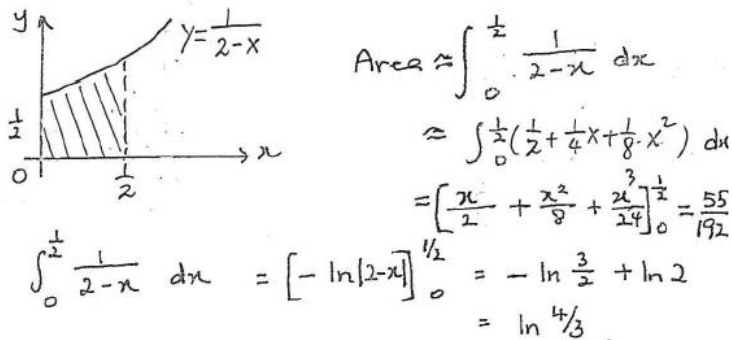
$$= \frac{1}{a} \left(1 - \frac{b}{a}x + \frac{(-1)(-2)}{2!} \left(\frac{b}{a}x\right)^2 + \dots\right)$$

$$= \frac{1}{a} - \frac{b}{a^2}x + \frac{b^2}{a^3}x^2 + \dots$$

$\therefore \frac{1}{a} - \frac{b}{a^2}x + \frac{b^2}{a^3}x^2 = \frac{1}{2} + \frac{1}{4}x + Cx^2$

$\Rightarrow a = 2, b = -1, C = \frac{1}{8}$

ii) $y = \frac{1}{2} + \frac{1}{4}x + \frac{1}{8}x^2 + \dots$



$\therefore \ln \frac{4}{3} \approx \frac{55}{192}$

4i) Let P_n be the proposition that

$$\frac{1}{x-1} - \frac{1}{x} - \frac{1}{x^2} - \dots - \frac{1}{x^n} = \frac{1}{x^n(x-1)}, \quad n \in \mathbb{N}^+, \quad x \notin \{0, 1\}$$

Consider P_1 :

$$\text{LHS} = \frac{1}{x-1} - \frac{1}{x} = \frac{x - (x-1)}{x(x-1)} = \frac{1}{x(x-1)}$$

$$\text{RHS} = \frac{1}{x(x-1)} = \text{LHS}.$$

$\therefore P_1$ is true.

Assume P_k is true for some $k \in \mathbb{N}^+$, i.e.

$$\frac{1}{x-1} - \frac{1}{x} - \frac{1}{x^2} - \dots - \frac{1}{x^k} = \frac{1}{x^k(x-1)}$$

Consider P_{k+1} ,

$$\text{LHS} = \frac{1}{x-1} - \frac{1}{x} - \dots - \frac{1}{x^k} - \frac{1}{x^{k+1}}$$

$$= \frac{1}{x^k(x-1)} - \frac{1}{x^{k+1}}$$

$$= \frac{x - (x-1)}{x^{k+1}(x-1)} = \frac{1}{x^{k+1}(x-1)} = \text{RHS}$$

$\therefore P_{k+1}$ is true.

Since P_1 is true $\left. \begin{array}{l} P_k \text{ true} \Rightarrow P_{k+1} \text{ true} \end{array} \right\} P_n \text{ is true for all } n \in \mathbb{N}^+.$

$$\text{ii) } \sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} \frac{1}{x^n(x-1)} = \sum_{n=1}^{\infty} \frac{1}{(x-1)} \left(\frac{1}{x}\right)^n$$

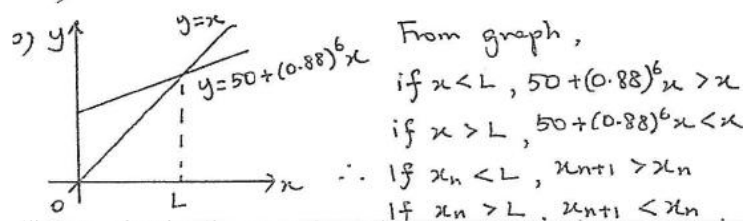
For S_{∞} to exist, $|\frac{1}{x}| < 1 \Rightarrow |x| > 1$

$\therefore x > 1$ or $x < -1$

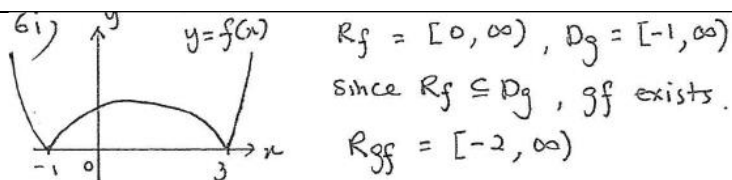
$$5c) u_n = 50(0.88)^n \Rightarrow u_6 = 23.2$$

$\therefore$ Amt of antibiotic remained = 23.2 units

$$\text{ii) a) } L = 50 + (0.88)^6 L \Rightarrow L = 93.4$$



Alternative Consider $x_{n+1} - x_n = 50 + 0.88^6 x_n - x_n$
 If $x_n < L$, $x_n < \frac{50}{1-0.88^6} \Rightarrow 50 + 0.88^6 x_n - x_n > 0$ i.e. $x_{n+1} > x_n$
 If $x_n > L$, $x_n > \frac{50}{1-0.88^6} \Rightarrow 50 + 0.88^6 x_n - x_n < 0$ i.e. $x_{n+1} < x_n$



ii) $f(-1) = f(3)$, $\therefore f$ is not 1-1.

largest $K = -1$.

iii) $f(x) = (x+1)(x-3)$ for $x \leq -1$.

$$\text{Let } y = x^2 - 2x - 3$$

$$= (x-1)^2 - 4$$

$$x = 1 \pm \sqrt{y+4}$$

$$= 1 - \sqrt{y+4} \quad \text{since } x \leq -1.$$

$$\therefore f^{-1}: x \mapsto 1 - \sqrt{x+4}, \quad x \geq 0.$$

7ai) $\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5 + \dots$

$$\ln(1-x) = -x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \frac{1}{4}x^4 - \frac{1}{5}x^5 + \dots$$

$$\ln\left(\frac{1+x}{1-x}\right) = \ln(1+x) - \ln(1-x)$$

$$= 2x + \frac{2}{3}x^3 + \frac{2}{5}x^5 + \dots$$

$$\sum_{r=0}^{\infty} \frac{1}{(2r+1)4^r} = 1 + \frac{1}{3 \cdot 4} + \frac{1}{5 \cdot 4^2} + \dots$$

$$= 2\left(\frac{1}{2}\right) + \frac{2}{3 \cdot 2^3} + \frac{2}{5 \cdot 2^5} + \dots$$

Let $x = \frac{1}{2}$, then $\sum_{r=0}^{\infty} \frac{1}{(2r+1)4^r} = \ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right) = \ln 3$.

b) $\sum_{r=37}^a (2r)^2 = 4 \left[\sum_{r=1}^a r^2 - \sum_{r=1}^{36} r^2 \right]$

$$= 4 \left[\frac{a}{6}(a+1)(2a+1) - \frac{36}{6}(36+1)(72+1) \right]$$

$$= \frac{2}{3}a(a+1)(2a+1) - 64824$$

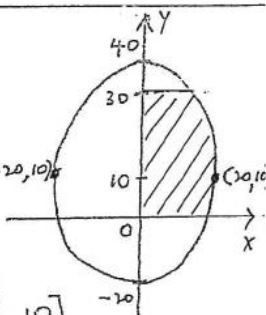
8i) $\frac{x^2}{20^2} + \frac{(y-10)^2}{30^2} = 1$

ii) $\text{Vol} = \int_0^h \pi x^2 dy$

$$= \pi \int_0^h 20^2 \left[1 - \frac{(y-10)^2}{30^2} \right] dy$$

$$= 400\pi \left[y - \frac{(y-10)^3}{900 \times 3} \right]_0^h$$

$$= \frac{4}{27}\pi [2700h - (h-10)^3 - 10]$$



$$8iii) \frac{dv}{dt} = 1000, \frac{dv}{dh} = \frac{4}{27} \pi [2700 - 3(h-10)^2]$$

$$\text{when } h = 15, \frac{dh}{dt} = \frac{dv}{dt} \times \left(\frac{dh}{dv} \right)$$

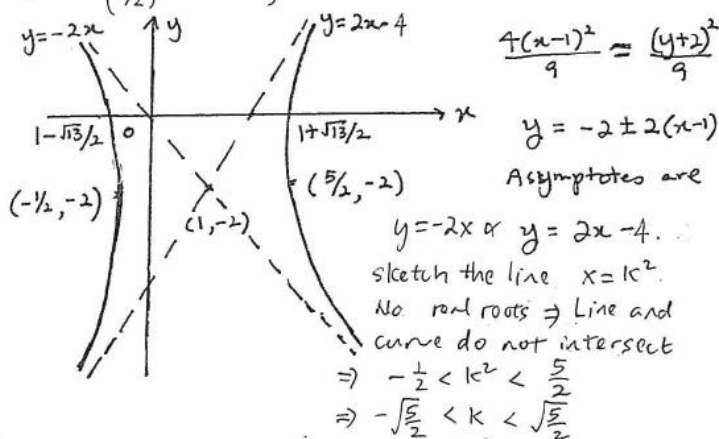
$$= 1000 \times \frac{27}{4\pi(2700 - 3 \times 5^2)}$$

$$= 0.819$$

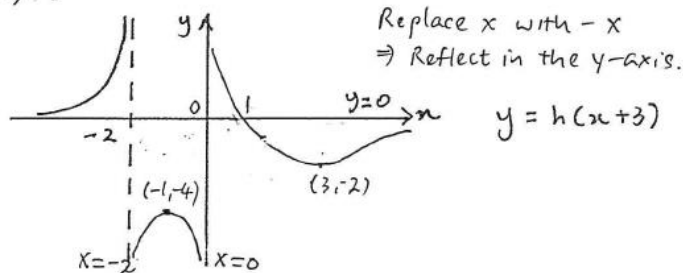
$\therefore$ the water level is increasing at $0.819 \text{ cm min}^{-1}$ when $h = 15$.

$\frac{dh}{dt}$ is min when $h-10=0$ i.e $h=10$

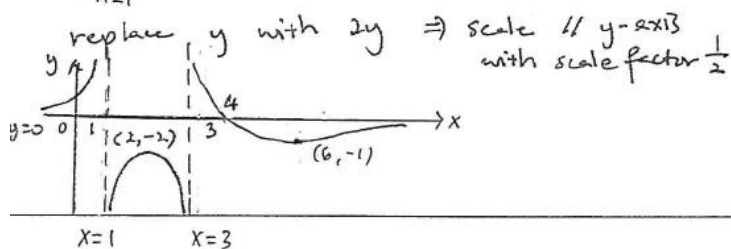
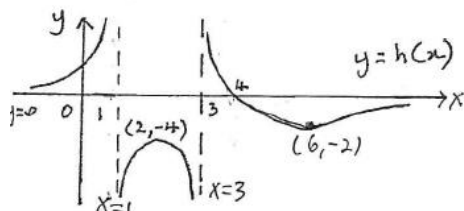
$$9a) \frac{(x-1)^2}{(\frac{3}{2})^2} - \frac{(y+2)^2}{3^2} = 1$$



b) i)



ii) replace x with $x-3$ for (i), $\Rightarrow$ translate 3 units in positive x -direction



$$10ai) \int \frac{1}{y} \ln y \, dy = (\ln y)^2 - \int (\ln y) \frac{1}{y} \, dy$$

$$\therefore 2 \int \frac{1}{y} \ln y \, dy = (\ln y)^2 + B$$

$$\int \frac{1}{y} \ln y \, dy = \frac{1}{2} (\ln y)^2 + C$$

$$ii) \frac{dy}{dx} = \frac{y}{\ln y}$$

$$\int \frac{1}{y} \ln y \, dy = \int 1 \, dx$$

$$x = \frac{1}{2} (\ln y)^2 + C$$

$$\ln y = \pm \sqrt{2(x-C)}$$

$$\text{Since } y > 1, \ln y > 0. \therefore \ln y = \sqrt{2(x-C)}$$

$$y = e^{\sqrt{2(x-C)}}$$

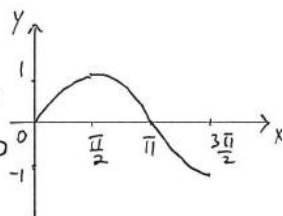
$$bi) \frac{d}{dx} [x \sinh(\ln x) - x \cosh(\ln x)]$$

$$= x \cosh(\ln x) \frac{1}{x} + \sinh(\ln x) + x \sinh(\ln x) \frac{1}{x} - \cosh(\ln x)$$

$$= 2 \sinh(\ln x)$$

$$ii) \frac{\pi}{2} \leq x \leq \pi, \sinh x \geq 0$$

$$\pi \leq x \leq \frac{3\pi}{2}, \sinh x \leq 0$$



$$\int_{e^{\pi/2}}^{e^{3\pi/2}} |\sin(\ln x)| \, dx$$

$$= \int_{e^{\pi/2}}^{e^{\pi}} \sinh(\ln x) \, dx - \int_{e^{\pi}}^{e^{3\pi/2}} \sinh(\ln x) \, dx$$

$$= \frac{1}{2} [x \sinh(\ln x) - x \cosh(\ln x)]_{e^{\pi/2}}^{e^{\pi}} - \frac{1}{2} [x \sinh(\ln x) - x \cosh(\ln x)]_{e^{\pi}}^{e^{3\pi/2}}$$

$$= \frac{1}{2} [e^{\pi} - e^{\pi/2} + e^{3\pi/2} + e^{\pi}] = e^{\pi} - \frac{1}{2} e^{\pi/2} + \frac{1}{2} e^{3\pi/2}$$

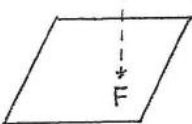
$$11) \underline{r} \cdot \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} = 7, \underline{r} = \begin{pmatrix} -8 \\ -2 \\ 8 \end{pmatrix} + t \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}, t \in \mathbb{R}$$

$$i) \vec{OA} = \begin{pmatrix} -8 \\ -2 \\ 8 \end{pmatrix} + 4 \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 20 \\ -6 \\ 16 \end{pmatrix}$$

$$\vec{OB} = \begin{pmatrix} -8 \\ -2 \\ 8 \end{pmatrix} + 2 \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 14 \\ -4 \\ 12 \end{pmatrix}$$

$$\vec{AB} = \begin{pmatrix} 14 \\ -4 \\ 12 \end{pmatrix} - \begin{pmatrix} 20 \\ -6 \\ 16 \end{pmatrix} = \begin{pmatrix} -6 \\ 2 \\ -4 \end{pmatrix}$$

$$\angle ABO = \cos^{-1} \left[\frac{\begin{pmatrix} -6 \\ 2 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 14 \\ -4 \\ 12 \end{pmatrix}}{\sqrt{56} \sqrt{356}} \right] = 172.5^\circ$$

(ii)  Equation of l_{AF} is
 $r = \begin{pmatrix} 20 \\ -6 \\ 16 \end{pmatrix} + s \begin{pmatrix} -2 \\ -3 \\ 6 \end{pmatrix}, s \in \mathbb{R}$
 At F, $\left[\begin{pmatrix} 20 \\ -6 \\ 16 \end{pmatrix} + s \begin{pmatrix} -2 \\ -3 \\ 6 \end{pmatrix} \right] \cdot \begin{pmatrix} -2 \\ -3 \\ 6 \end{pmatrix} = 7$

$$\Rightarrow 154 + 49s = 7$$

$$\Rightarrow s = -3$$

$$\therefore \vec{OF} = \begin{pmatrix} 20 \\ -6 \\ 16 \end{pmatrix} - 3 \begin{pmatrix} -2 \\ -3 \\ 6 \end{pmatrix} = \begin{pmatrix} 14 \\ 3 \\ -2 \end{pmatrix}$$

coordinates of F is (14, 3, -2).

(iii) $x=0 \Rightarrow r \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0$

Let θ be the angle b/w π & $x=0$

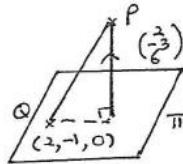
$$\cos \theta = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot \frac{\begin{pmatrix} -2 \\ -3 \\ 6 \end{pmatrix}}{7} = \frac{2}{7}$$

$$\theta = 73.4^\circ \therefore \text{The acute } \angle \text{ is } 73.4^\circ$$

(iv) $\vec{OP} = \begin{pmatrix} 8 \\ -2 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$

$Q(2, -1, 0)$ is a pt in π .

$$\vec{PQ} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} - \begin{pmatrix} 8+3\lambda \\ -2-\lambda \\ 8+2\lambda \end{pmatrix} = \begin{pmatrix} -6-3\lambda \\ 1+\lambda \\ -8-2\lambda \end{pmatrix}$$



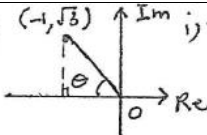
$$\text{h dist} = |\vec{PQ} \cdot \hat{n}| = \left| \begin{pmatrix} -6-3\lambda \\ 1+\lambda \\ -8-2\lambda \end{pmatrix} \cdot \frac{\begin{pmatrix} -2 \\ -3 \\ 6 \end{pmatrix}}{7} \right|$$

$$= \frac{1}{7} |63 + 21\lambda| = |9 + 3\lambda|$$

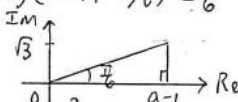
$$|9 + 3\lambda| < 3 \Rightarrow -3 < 3\lambda + 9 < 3$$

$$\Rightarrow -4 < \lambda < -2$$

$$\therefore \{ \lambda \in \mathbb{R} : -4 < \lambda < -2 \}$$

12a) $(-1, \sqrt{3})$  $j, \tan \theta = \sqrt{3} \Rightarrow \theta = \pi/3$
 $\therefore \arg(z) = 2\pi/3$

ii) $\arg(z(z+a)) = \arg z + \arg(z+a) = \frac{5\pi}{6}$

$\Rightarrow \arg(z+a) = \frac{5\pi}{6} - \frac{2\pi}{3} = \frac{\pi}{6}$
 $\Rightarrow \arg(a-1 + j\sqrt{3}) = \frac{\pi}{6}$  $\therefore \tan \frac{\pi}{6} = \frac{\sqrt{3}}{a-1} \Rightarrow \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{a-1}$
 $\Rightarrow a = 4$

b) $iz^3 = 3-3i \Rightarrow z^3 = \frac{3-3i}{i} = -3-3i$

$z^3 = \sqrt{18} e^{i(-\frac{3\pi}{4} + 2k\pi)}$, $k = 0, \pm 1$

$z = 18^{1/6} e^{i(-\frac{\pi}{4} + \frac{2k}{3}\pi)} = 18^{1/6} e^{i(\frac{8k-3}{12})\pi}$

$= 18^{1/6} e^{i(-\pi/4)}, 18^{1/6} e^{i(5\pi/12)}, 18^{1/6} e^{i(-11\pi/12)}$

$-w^3 = 3-3i$

$\Rightarrow i(-iw)^3 = 3-3i$

$-iw = z \Rightarrow w = \frac{z}{-i} = iz = e^{i\pi/2} \cdot z$

$\therefore w = 18^{1/6} e^{i(\pi/4)}, 18^{1/6} e^{i(11\pi/12)}, 18^{1/6} e^{i(5\pi/12)}$

Alternative

$-w^3 = 3-3i$

$w^3 = -3+3i$

$= (-3-3i)^*$

$= (z^3)^*$

$= (z^*)^3$

$\therefore w = z^*$

$= 18^{1/6} e^{i\pi/4}, 18^{1/6} e^{-i5\pi/12}, 18^{1/6} e^{i11\pi/12}$